Review addendum: PR #1164

Date: 2026-10-29. Certify sqrt2 Frobenius factor swap at j1728.

Original author head: 37425b94bda34eb1e411b3d0e981a7e83623e6a5. The original study and every frozen source, protocol, result, failed attempt and timing cap are preserved. This addendum takes precedence over unqualified experimental-status language in the archived report.

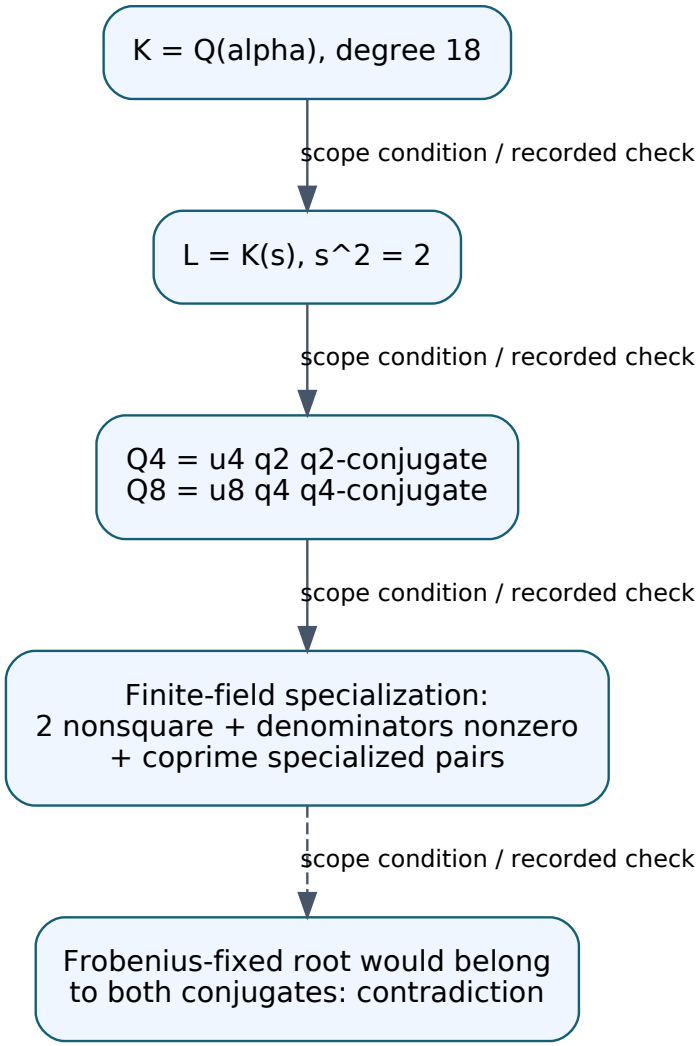
Recorded finding and its scope

The exact quartic and octic factors of the source-j=1728 C–E resolvent each split over K(√2) into a pair exchanged by √2↦−√2: two irreducible quadratics and two irreducible quartics, respectively. Each pair is coprime over characteristic zero. At a rational C point modulo a prime where 2 is nonsquare, the same factor exchange becomes Frobenius on F_{p^2}. When the specialized pair remains coprime, a root fixed by Frobenius would have to lie in both factors, which is impossible. This gives a conditional arithmetic explanation for the zero rational C–E rows in the tested $p \equiv 11 \pmod{24}$ class.

The paragraph above describes the retained certificate record, not a new experiment. Source: experiments/hecke-432-j1728-sqrt2-split-20261026/RESULT.md (../../experiments/hecke-432-j1728-sqrt2-split-20261026/RESULT.md). This review treats publication of protocol and results together as retrospective/exploratory unless an independent earlier publication witness is supplied. An archived PASS_PROSPECTIVE status is a literal historical label, not evidence of pre-observation registration. In particular, PR #1164's six-row holdout and PR #1190's 62/62 correctness checks retain their numerical results with this qualification.

Evidence-linked explanation

Frobenius exchanges coprime conjugate factors. Symbols C, B, E and K denote the frozen coordinate models or coefficient field, not newly registered subgroup instances. This diagram records algebra/certificate dependencies; it does not introduce a verified isogeny edge, EC1 identity, ISO1 route, or ECDLP transport claim.



Editable source: 1164.dot (1164.dot). Report and visual: 1164.pdf (1164.pdf).

Replay and proof map

The exact manifest binds 17 files in this study. The shared project pins Lean leanprover/lean4:v4.35.0-rc4 and Mathlib 021c46bdd042f67b538b9f9871d556a93713a9a9. All 37 original modules, 113 requested declaration reports and 142 noninternal theorems passed the integrated proof replay; this does not formalize every concrete Sage/PARI field or ideal interpretation. The source manifest and proof map are tools/ci/quartic_certificates.json (../../tools/ci/quartic_certificates.json) and the original integration report (../ci-repair-quartic-20261027.md).

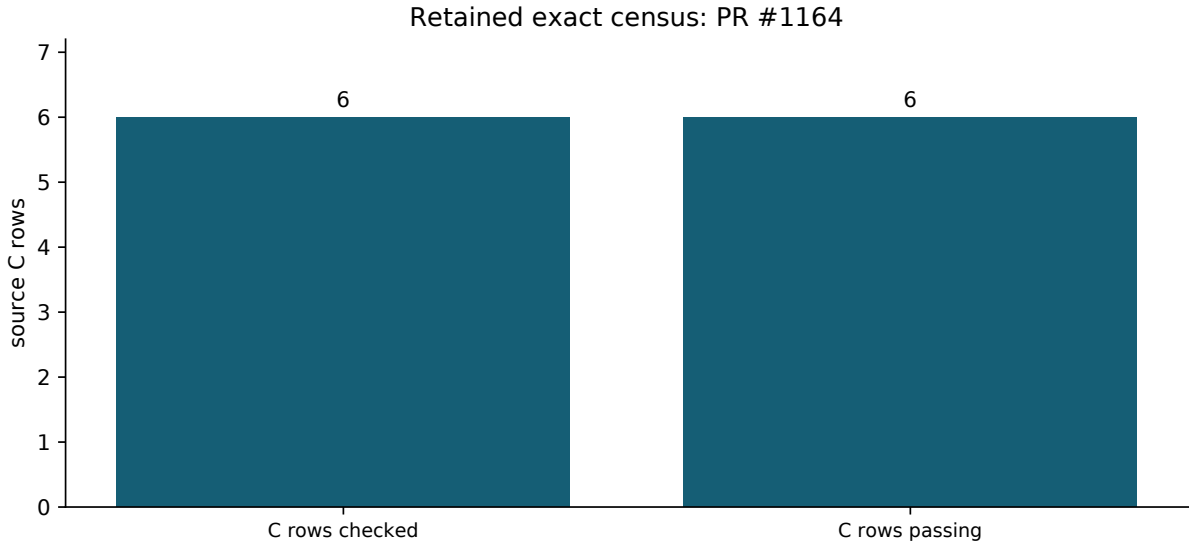
This study's original modules: experiments/hecke-432-j1728-sqrt2-split-20261026/formal/FrobeniusSwapGate.lean.

A new portable replay ran all 42 original integer auditors successfully in a disposable canonical-layout container. The checkout was read-only and all 1,404 original input hashes were verified before and after replay. See portable replay evidence (portable-replay/results.json). The separate field-index successor runner uses new output directories, unchanged source/protocol bytes, original timeout limits and the checked Sage launcher. Preparation and I/O controls are not a new Sage experiment.

Remaining obligations and graph review

Arithmetic table certificates and universal ring/permutation identities retain their precise hypotheses. Concrete number-field encodings, integral bases, ideals, Hensel lifting and Frobenius interpretation require their own bridges. No security, speedup, prospective-validation or general impossibility claim is added.

The candidate catalog, volcano-IC figure set and paper figures were checked for affected links. This is a review and portability correction with no new surviving graphable mathematical fact; canonical graphs remain unchanged. The precise checked-path inventory is graph-review.json (graph-review.json).

Retained exact counts

Six saved C rows at primes 8123, 8147 and 8171. Retrospective verification. Units: source C rows. These are exact finite-census counts with no statistical error bars; extrapolation beyond the saved domain is not established. No new timings were collected. Data: experiments/hecke-432-j1728-sqrt2-split-20261026/holdout_result.json (../../experiments/hecke-432-j1728-sqrt2-split-20261026/holdout_result.json), SHA-256 1bac73fbce6500c9d6a93acc6ad0af632b0889b16b04e61e8e99af9f0a13c459; plotted values (quantitative-sources.json).